uPAR (PLAUR) as a Senescence-Enriched Surface Marker and Senolytic Target — Mechanistic Hypothesis Search

Hypothesis ID: `upar_senescence_surface_marker_target` **Module:** Conserved Cellular Senescence Module **Status in KB:** EMERGING **Search date:** 2026-10-18

Executive Judgment

Verdict: PARTIALLY SUPPORTED — the hypothesis is correct in its therapeutic core but overstated in its specificity claim. The current literature cleanly confirms the KB's own framing that uPAR is **senescence-enriched, not senescence-specific**.

Three pillars of the hypothesis are well-supported. First, PLAUR is a **verified member of the SenMayo senescence gene signature** (MSigDB `SAUL_SEN_MAYO`, M45803) and encodes a genuinely surface-accessible, GPI-anchored protein. Second, **uPAR-positive cells accumulate with age in parallel with SA-β-gal-positive cells**, and clearing them is causally beneficial: anti-uPAR CAR T cells (and independently-developed BiTEs, chimeric "matchmaker" peptides, and uPAR-targeted nanoparticles) reverse metabolic, intestinal, hepatic-fibrotic and arthritic aging phenotypes across multiple independent mouse studies. Third, **soluble uPAR (suPAR) tracks accelerated biological aging in a population-representative human cohort** (Dunedin Study), providing a human aging correlate.

However, the specificity premise — that uPAR "marks the age-accumulating senescent-cell population well enough to serve as a therapeutic handle" — is only **quantitative, not categorical**. Human Protein Atlas data show uPAR is constitutively high on normal neutrophils, monocytes, dendritic cells, macrophages and endothelium, is "Detected in all" tissues, and is classified "Cell type enhanced" (never "Cell type specific"). No candidate senescence surface marker in an 11-gene panel was cell-type-specific at baseline, so **any specificity must come from senescence-driven upregulation above a non-zero baseline rather than from a senescence-restricted on/off expression pattern**. On-target toxicity is already documented (a uPAR T-cell engager caused dose-dependent hepatotoxicity via attack

on hepatic endothelial cells). The human evidence is thin: a soluble-suPAR aging correlate and a single rheumatoid-arthritis scRNA dataset, with **zero registered human senolytic/uPAR CAR-T trials** on ClinicalTrials.gov as of the search date. Human genetics (Open Targets) links PLAUR to age-related cardiovascular/inflammatory disease and leukocyte count — consistent with its myeloid/vascular biology — but shows **no direct genetic association with aging, longevity, or cellular-senescence terms**.

The most important caveat for curators: the therapeutic efficacy evidence (strong, causal, multi-tissue, mouse) and the specificity evidence (weak — uPAR is broadly expressed) are not in conflict; they coexist because senolytic efficacy does not require a perfectly specific marker, only a favorable therapeutic window. The hypothesis should retain **EMERGING** status, with its specificity qualifier preserved and strengthened, and with an explicit note that human translation is unproven.

Key Findings

F001 — uPAR is constitutively high on normal myeloid cells (specificity ceiling)

Human Protein Atlas consensus RNA data (retrieved 2026-10-18 via the HPA search/download API, ENSG00000011422) classify **PLAUR as "Tissue enriched" but "Detected in all" tissues**; its single-cell class is "Cell type enhanced" (explicitly *not* "Cell type specific"), and its blood class is "Immune cell enhanced." Ranked single-cell expression (nCPM) is dominated by normal myeloid lineage: Neutrophils 7137.8, Epididymal basal 3927.4, monocytes 2786.3, Ocular epithelial 2333.0, conventional dendritic cells (cDC) 1115.4, Endometrial secretory 874.8, Macrophages 787.2. The top expressers are overwhelmingly normal, non-senescent cells.

This baseline expression has direct therapeutic consequences. A uPAR-targeting T-cell engager (GFD-CD3) produced **dose-dependent hepatotoxicity through T cell-mediated attack on hepatic endothelial cells** (PMID: 42806572) — an on-target, off-senescent-cell effect. Independently, uPAR "is broadly expressed in solid tumors enriched for TP53 and RAS pathway mutations" (PMID: 41916312), confirming that uPAR surface expression is broad rather than restricted to senescent cells. **This finding is the mechanistic basis for the "enriched, not specific" caveat.**

F002 — PLAUR is a verified SenMayo member, but embedded in a secretory inflammatory program

The authoritative SenMayo set (`SAUL_SEN_MAYO`, systematic name M45803, source PMID: 35974106, Saul et al. 2022 *Nat Commun*, Supplementary Data 1) was retrieved from MSigDB on 2026-10-18. A programmatic membership check confirmed `'PLAUR' in geneSymbols = True` (index 99 of 124 genes). Critically, the set also contains the full plasminogen-activation/fibrinolysis axis (PLAT, PLAU, PLAUR, SERPINE1) and is dominated by secreted SASP factors (CCL/CXCL chemokines, IL1A/IL1B/IL6/CXCL8, MMP1/2/3/9/13, IGFBP family, VEGFA, TNF).

The interpretation is nuanced: PLAUR is the **only canonical surface-anchored candidate among the SenMayo genes**, which is precisely why it is attractive as a therapeutic handle — but it is embedded in a *secretory* inflammatory program. Its presence in SenMayo reflects SASP/inflammation biology as much as a senescence-specific surface identity. This aligns with its known role as a GPI-anchored receptor in the fibrinolysis cascade and leukocyte activation.

F003 — Human suPAR tracks accelerated biological aging (supportive but indirect)

In the Dunedin Study, a population-representative 1972–73 New Zealand birth cohort (n=1037; suPAR measured at ages 38 and 45, 875/997 living participants at 45), **elevated plasma suPAR was associated with accelerated pace of biological aging across multiple organ systems, older facial appearance, older brain age, and greater decline in physical/cognitive function**; improvements in health habits blunted the suPAR rise (PMID: 32766674, Rasmussen et al. 2021). Regressions adjusted for sex, smoking, CRP, and current health conditions.

This is the primary human cohort evidence cited in the seed hypothesis. It is **supportive but indirect**: it measures *soluble* uPAR (a cleaved/shed circulating form, an inflammation biomarker) rather than *membrane* uPAR on senescent cells, and correlation with aging pace does not establish that suPAR originates from senescent cells or that membrane uPAR marks them specifically.

F004 — uPAR is one of several competing senescent-cell surface markers; off-target toxicity is a recurring theme

uPAR is used as a senescent-cell handle beyond CAR T: a chimeric "matchmaker" peptide binds uPAR to drive immune clearance of senescent cells in liver fibrosis, lung injury, cancer and natural aging (PMID: 39623223, 2025), explicitly noting that senolytics are "limited by off-target toxicity." Competing/alternative senescent-cell surface markers under active development

include **GPNMB** (engineered senolytic EVs, PMID: 38794820), **NKG2D ligands** (anti-NKG2D CAR T, PMID: 41476987), **TSPAN8** (senescent CAFs, PMID: 41577205), and **TM4SF1** (senescent pancreatic β-cells, PMID: 42425712). A 2025 review concludes that key unresolved challenges are "the identification of specific and universal senescence markers, potential off-target effects, and long-term safety concerns," with a "significant translational gap" (PMID: 41476987).

The field-level picture is that uPAR is a **leading but not unique** surface handle, and that marker specificity plus off-target toxicity are acknowledged, unsolved problems across all such markers.

F005 — No candidate senescence surface marker is cell-type-specific at baseline

An HPA consensus RNA comparison (2026-10-18) across 11 senescence-associated genes plus a specificity positive control found that **all of PLAUR, CDKN2A, CDKN1A, GPNMB, DPP4, B2M, TM4SF1, SERPINE1, LMNB1, TSPAN8, and NCAM1 are classified "Cell type enhanced" (not "Cell type specific")**; only the melanocyte gene MLANA (positive control) is "Cell type enriched/specific." Among the panel, PLAUR carries the **highest absolute immune-cell baseline expression** (Neutrophils 7138, monocytes 2786 nCPM) and is "Detected in all" tissues.

The implication is structural: because no senescence surface marker is specific at baseline, **specificity for senolysis can only come from senescence-driven upregulation above baseline**, not from a clean on/off switch. For uPAR, the high myeloid baseline makes the required fold-change discrimination harder than for lower-baseline markers.

F006 — Strongest direct evidence: anti-uPAR CAR T causally reverses age-related pathologies in mice (tissue-dependent target identity)

This is the strongest pillar of the hypothesis. Convergent, interventional mouse evidence that clearing uPAR+ cells reverses age-related dysfunction:

1. **Proof-of-concept:** senolytic anti-uPAR CAR T cells safely reverse senescence-associated pathologies (PMID: 32555459, Amor et al. 2020 *Nature*).
2. **Durable efficacy:** a *single* administration improves exercise capacity in physiological aging and ameliorates metabolic dysfunction/glucose tolerance in aged and high-fat-diet mice — "a single administration of these senolytic CAR T cells is sufficient to achieve long-term therapeutic and preventive effects" (PMID: 38267706, Amor et al. 2024).
3. **Second tissue:** "both therapeutic and prophylactic treatment with anti-uPAR CAR T cells improved barrier function, regenerative capacity, inflammation, mucosal immune function

and microbiome composition in aged mice" — with the accumulating target cells being "mostly epithelial uPAR-positive cells in the aging gut" (PMID: 41291258, 2026).

4. **In-vivo manufacturing + human signal:** mRNA/LNP uPAR CAR-T alleviates liver fibrosis and rheumatoid arthritis, and "Single-cell sequencing in humans confirms uPAR's relevance to senescence and inflammation in RA" (PMID: 40602406, 2025).

Two points are curation-critical. First, the efficacy is **causal and reproducible across independent groups and tissues** (metabolic, intestinal, hepatic, arthritic). Second, **the uPAR+ target cell identity is tissue-dependent** — endothelial/myeloid/preadipocyte in prior work, "mostly epithelial" in the aging gut — reinforcing that uPAR marks a heterogeneous, context-dependent population rather than a single senescent cell type.

F007 — Source-level gap: no registered human senolytic/uPAR CAR-T trial

ClinicalTrials.gov API v2 was searched 2026-10-18 (query.term, countTotal=true). Results:

Query	Total hits	Nature of hits
`senolytic`	31	all small-molecule senolytics (OA/PAD/vascular/pulmonary aging); none CAR-T
`senescent cell CAR T`	72	all oncology CAR-T; none senolytic or uPAR
`senolytic CAR`	0	—
`uPAR CAR`	0	—
`PLAUR CAR`	0	—
`urokinase CAR T`	0	—
`uPAR senolytic`	0	—

API functionality was confirmed by non-zero calibration queries. **As of the search date there is no registered human trial of a uPAR-directed senolytic CAR-T (or any senolytic CAR-T).** The search log is saved in the artifact bundle (`artifacts/data/clinicaltrials_search_log.json`). This is a genuine, verified source-level absence, not an unchecked assumption.

F008 — Human genetics (Open Targets): PLAUR tracks cardiovascular/ inflammatory disease and leukocyte count, not senescence

Open Targets Platform GraphQL (accessed 2026-10-18; PLAUR/ENSG00000011422; 1616 associated diseases) returned highest genetic-association scores for: **abdominal aortic aneurysm 0.813, "Decreased total leukocyte count" 0.747, myocardial infarction 0.604, angina pectoris 0.597, aortic aneurysm 0.597, coronary artery disorder 0.592**; neurodegenerative/Parkinson/Alzheimer terms appear only at the overall-score level. There was **no direct genetic association with aging/longevity/cellular-senescence terms**. The association profile is exactly what uPAR's known myeloid/vascular/fibrinolytic biology predicts, and it does *not* independently nominate uPAR as a senescence gene. Artifact: `data/opentargets_plaur_associations.json`.

Mechanistic Model / Interpretation

The hypothesis implies the following causal chain. Strength of support is annotated at each link.

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Aging / oncogenic stress / therapy-induced damage
      │  [STRONG: senescence biology is well established]
      ▼
Cellular senescence program (p16/p21, SA-β-gal, SASP)
      │  [STRONG: PLAUR verified in SenMayo (M45803, F002)]
      ▼
Upregulation of surface uPAR on (a fraction of) senescent cells
      │  [MODERATE: uPAR+ cells accumulate with age parallel to SA-β-gal (seed <a href="https
      │   BUT high non-senescent baseline on myeloid/endothelial cells (F001, F005)]
      ▼
uPAR+ cells accumulate in aged tissue (tissue-dependent identity:
endothelial / myeloid / epithelial / preadipocyte)
      │  [STRONG in mouse (F006); target identity shifts by tissue]
      ▼
Anti-uPAR CAR-T / BiTE / peptide / nanoparticle clears uPAR+ cells
      │  [STRONG causal mouse evidence, multi-tissue, multi-modality (F006, F004)]
      │  [RISK: on-target clearance of normal uPAR+ cells → hepatotoxicity (F001)]
      ▼
Reversal of age-related tissue dysfunction
      │  [STRONG in mouse: metabolic, intestinal, hepatic-fibrotic, arthritic (F006)]
      ▼
Clinical benefit in humans
         [UNPROVEN: no human trial (F007); human data = suPAR aging
          correlate (F003) + 1 RA scRNA dataset (F006); genetics
          points to CVD/leukocyte biology, not senescence (F008)]
```

Where the literature is strong: the two termini — senescence biology upstream and causal therapeutic reversal in mice downstream. Clearing uPAR+ cells demonstrably reverses aging phenotypes across four tissues and at least four targeting modalities.

Where links are inferred or weak: (1) the *specificity* of the senescence→surface-uPAR step — uPAR is upregulated on *a fraction* of senescent cells but is also constitutively high on normal cells, so the marker separates senescent from non-senescent cells only quantitatively; (2) the human translation terminus — essentially unvalidated, resting on a soluble-biomarker correlate and one scRNA dataset.

The reconciling insight: therapeutic efficacy and imperfect specificity coexist because senolysis exploits a *therapeutic window* (senescent cells may be more abundant, more uPAR-high, or more vulnerable), not a binary marker. The mouse efficacy data do not require — and do not demonstrate — clean specificity.

Evidence Matrix

Citation	Evidence type	Stance	Mechanistic claim tested	Key finding	Context	C
PMID: 38267706	Model organism	**Support**	uPAR+ cells accumulate with age & clearing them reverses metabolic dysfunction	Single-dose senolytic CAR-T gives durable therapeutic+preventive benefit; uPAR+ co-accumulates with SA-β-gal+	Aged / HFD mice	
PMID: 32555459	Model organism	**Support**	Senolytic anti-uPAR CAR-T reverses senescence pathologies	Proof-of-concept: safe reversal of senescence-associated pathologies	Atherosclerosis, liver fibrosis (mouse)	
PMID: 41291258	Model organism	**Support**	Clearing uPAR+ cells restores tissue fitness in a second organ	Improved intestinal barrier, regeneration, inflammation, microbiome; target cells "mostly epithelial"	Aged mouse gut	
PMID: 40602406	Model organism + human scRNA	**Support / qualify**	In-vivo uPAR CAR-T treats fibrosis/RA; human relevance	Alleviates liver fibrosis & RA; human scRNA confirms uPAR relevance to senescence/ inflammation in RA	Mouse + human RA	
PMID: 32766674	Human clinical (cohort)	**Support (indirect)**	Soluble uPAR correlates with aging pace	Elevated suPAR → accelerated biological aging across organ systems	Dunedin cohort, n=1037	
		Support			Multi-model	

Citation	Evidence type	Stance	Mechanistic claim tested	Key finding	Context
PMID: 39623223	In vitro / model organism		uPAR usable as senescent-cell handle beyond CAR-T	Chimeric peptide binds uPAR to clear senescent cells in fibrosis/injury/aging	
PMID: 42806572	Model organism (mouse + NHP)	**Refute / qualify**	uPAR targeting is specific & safe	uPAR BiTE caused dose-dependent hepatotoxicity via attack on hepatic endothelial cells	Mouse, non-human primate
PMID: 41916312	Model organism / computational	**Qualify**	uPAR expression is senescence-restricted	uPAR "broadly expressed in solid tumors" (TP53/RAS enriched)	Solid tumors
HPA consensus RNA (F001, F005)	Computational (database)	**Qualify / refute specificity**	uPAR is a senescence-specific surface tag	PLAUR "Cell type enhanced," "Detected in all," highest myeloid baseline of 11-gene panel; no marker is specific	Human, cross-tissue
Open Targets (F008)	Computational (genetics)	**Qualify**	PLAUR is genetically a senescence gene	Top associations = AAA, leukocyte count, MI, angina; no aging/ senescence terms	Human genetics
PMID: 41476987	Review	**Qualify**	Senescence markers are specific & safe	Unresolved: "specific and universal senescence markers, off-target effects, long-term safety"; "significant translational gap"	Field review

Citation	Evidence type	Stance	Mechanistic claim tested	Key finding	Context
PMID: 38794820	In vitro / model organism	**Competing**	uPAR is the surface handle for senolysis	GPNMB is an alternative senescent-cell surface marker (engineered EVs)	Tumor microenvironment
PMID: 42425712	Model organism	**Competing**	uPAR marks senescent β-cells	TM4SF1 identified as surface marker of senescent β-cells (vs uPAR)	Pancreatic β-cells / T2DM
PMID: 41577205	Review / model organism	**Competing**	uPAR is the universal senescent-stroma marker	TSPAN8 marks senescent CAFs	Tumor stroma
PMID: 42442548	Model organism	**Support**	uPAR-targeted delivery reduces senescent-adipocyte burden	uPAR-targeted D+Q nanoparticles improve insulin sensitivity in HFD obese mice	Aged/obese adipose
PMID: 42312503	Review	**Qualify**	uPAR CAR-T is clinically ready	uPAR senolytic CAR-T "remain at a preclinical stage" for lymphoma TIS	NHL therapy-induced senescence

Data and Tool Use

Source	Accession / URI	Access status	Query / filters	Relevance	Artifact
Human Protein Atlas (consensus RNA, single-cell, blood)	ENSG00000011422 (PLAUR) + 11-gene panel	**Accessed** (2026-10-18, search/ download API)	Expression class + ranked nCPM across cell types/ tissues	Directly relevant: baseline specificity of the marker	HPA tables (F001
MSigDB — SenMayo	`SAUL_SEN_MAYO` / M45803 (source P 35974106)	**Accessed** (2026-10-18)	`'PLAUR' in geneSymbols` membership check	Directly relevant: verifies seed claim of SenMayo membership	Gene list + memb
ClinicalTrials.gov API v2	api.clinicaltrials.gov/ v2	**Accessed; negative result verified** (2026-10-18)	7 query.term searches (table in F007), countTotal=true	Directly relevant: human trial landscape	`artifacts/data clinicaltrials` (F007)
Open Targets Platform	GraphQL; ENSG00000011422	**Accessed** (2026-10-18)	PLAUR associated diseases, ranked by genetic-association score	Directly relevant: human genetic nomination	`data/ opentargets_pl` (F008)
PubMed (via search_pubmed)	33 papers reviewed	**Accessed**	uPAR/PLAUR senescence, senolytic CAR-T, SenMayo, suPAR aging	Directly relevant: primary + review literature	Literature list (th

Analysis inventory (input → method → output):

1. **HPA specificity analysis (F001/F005)** — Input: HPA consensus RNA + single-cell nCPM for PLAUR and an 11-gene senescence panel + MLANA control. Method: retrieve expression-class annotations and rank absolute nCPM. Output: PLAUR = "Cell type enhanced," "Detected in all," highest myeloid baseline; no panel gene specific. **Outcome: succeeded.**
2. **SenMayo membership check (F002)** — Input: MSigDB `SAUL_SEN_MAYO` gene list (124/125 symbols). Method: programmatic string membership. Output: `True`, index 99. **Outcome: succeeded.**
3. **ClinicalTrials.gov trial census (F007)** — Input: 7 term queries. Method: API v2 countTotal calls with calibration controls. Output: all uPAR/senolytic-CAR queries = 0; small-molecule/oncology controls non-zero. **Outcome: succeeded (verified negative).**
4. **Open Targets genetics (F008)** — Input: PLAUR gene ID. Method: GraphQL associated-diseases query, sort by score. Output: top terms AAA/leukocyte/MI/angina; no senescence terms. **Outcome: succeeded.**

No patient-level or controlled-access data were used. No analysis was proposed-but-not-run; no fallback from a failed computational analysis to literature synthesis occurred. The HPA/Open Targets analyses rely on *bulk/enhanced* expression classes and are **not resolved by senescence state** — they bound baseline specificity but cannot measure senescence-induced fold-change, which is the actual discriminator; this is a stated limitation.

Limitations and Knowledge Gaps

1. Senescence-state-resolved surface uPAR quantification is missing. All specificity evidence here (HPA, Open Targets) reflects baseline/bulk expression in normal cells, not the uPAR surface density on *senescent* vs *non-senescent* cells of the same type. *Why it matters:* the therapeutic window depends on the fold-change, not the baseline. *What was checked:* HPA expression classes (no senescence axis available). *Resolution:* paired flow cytometry / CyTOF of surface uPAR on SA-β-gal+ vs SA-β-gal− cells across tissues.

2. Human translation is essentially unvalidated. *What was checked:* ClinicalTrials.gov (7 queries, 2026-10-18) → zero senolytic/uPAR CAR-T trials; human data limited to the Dunedin suPAR correlate (soluble, not membrane) and one RA scRNA dataset. *Resolution:* a first-in-human dose-finding senolytic CAR-T trial with uPAR surface-expression biomarker stratification.

3. On-target toxicity against normal uPAR+ cells is documented but not bounded. Hepatic-endothelial injury occurred with a uPAR BiTE (PMID: 42806572). *Why it matters:* defines the safety ceiling. *Resolution:* dose-response toxicity studies with serum aminotransferase monitoring (already proposed as a safety readout in that paper) and transient/controllable CAR designs.

4. The suPAR→membrane-uPAR→senescent-cell link is inferred, not demonstrated. Soluble suPAR correlating with aging pace does not establish that it is shed from senescent cells. *Resolution:* source-tissue deconvolution of plasma suPAR; longitudinal suPAR vs tissue senescent-cell burden.

5. Genetic nomination is absent. Open Targets shows PLAUR's human genetics point to CVD/leukocyte biology, with no aging/senescence terms. *Why it matters:* no Mendelian-randomization-style human causal support that uPAR biology drives senescence/aging. *Resolution:* MR of PLAUR variants against aging-pace / senescence-burden phenotypes.

6. Tissue-dependent target identity is unresolved for specificity. uPAR+ cells are endothelial/myeloid in some tissues and "mostly epithelial" in the aging gut (PMID: 41291258). *Resolution:* cross-tissue single-cell atlas of uPAR+ cells co-stained for canonical senescence markers.

Alternative Models

Alternative hypothesis	Relationship to seed	Rationale
Competing surface markers (GPNMB [PMID 38794820], NKG2DL [PMID 41476987], TSPAN8 [PMID 41577205], TM4SF1 [PMID 42425712])	**Alternative/ parallel**	Each is a surface marker for senescence in a specific tissue; several may be more tissue-specific than uPAR. uPAR is a leading but not unique handle.
uPAR as SASP/ inflammation marker, not senescence-surface tag	**Reframing/ competing**	PLAUR sits inside the SenMayo *secretory* program (F002) and its human genetics/ expression track myeloid-vascular inflammation (F001, F008). Benefit from clearing uPAR+ cells may be driven by removing inflammatory myeloid/endothelial cells (inflamm-aging), not canonically senescent cells.
Therapeutic-window model (quantitative enrichment)	**Refinement of the seed**	Senolytic efficacy requires only that uPAR+ burden/density is higher in diseased/aged tissue, not binary specificity. Best reconciles strong efficacy (F006) with weak specificity (F001/F005).
Small-molecule senolytics (D+Q, navitoclax, fisetin)	**Parallel mechanism**	Achieve senolysis via pro-survival-pathway inhibition without any surface marker; the dominant clinical modality (all 31 ClinicalTrials.gov senolytic trials).
Soluble suPAR as a bystander aging biomarker	**Downstream consequence**	suPAR may index systemic inflammation/organ damage rather than cause or specifically mark tissue senescence (F003).

Discriminating Tests

1. **Senescence-resolved surface-uPAR cytometry (specificity test).** Sample: multiple aged mouse tissues + human tissue where available. Perturbation: none (observational). Readout: surface uPAR density on SA-β-gal+/p16+ vs matched non-senescent cells of each type. *Expected if seed is right:* senescent cells carry significantly higher surface uPAR than same-type non-senescent cells; *if the alternative (myeloid-inflammation) model holds:* uPAR density is driven by lineage, not senescence state.
2. **Head-to-head marker comparison for therapeutic index.** Compare anti-uPAR vs anti-GPNMB vs anti-NKG2DL CAR-T/EV for senolytic efficacy and on-target toxicity in the same aged-mouse models. *Expected:* ranks markers by therapeutic window; informs which is most tissue-selective.
3. **First-in-human biomarker-stratified senolytic CAR-T trial.** Stratify patients (e.g., metabolic dysfunction, fibrosis, RA) by tissue uPAR+ senescent-cell burden; primary safety endpoint includes serum aminotransferases (hepatic-endothelial toxicity monitor per PMID: 42806572). *Expected:* establishes human efficacy/safety and whether uPAR+ burden predicts response.
4. **Mendelian randomization of PLAUR variants vs aging-pace phenotypes.** Using the genetic instruments already catalogued in Open Targets (F008). *Expected if seed is right:* PLAUR variants causally associate with senescence/aging-pace outcomes; *current null genetic signal predicts a negative/weak result*, which would favor the SASP-biomarker reframing.
5. **Plasma suPAR source deconvolution.** Determine whether circulating suPAR derives from senescent cells vs activated myeloid cells, via tissue-conditional PLAUR knockout + plasma suPAR measurement. *Expected:* clarifies whether the human suPAR-aging correlate (F003) reflects senescence biology.

Curation Leads (require curator verification)

Candidate evidence references & snippets to verify: - PMID: 42806572 — "high-dose GFD-CD3 predominantly induced hepatotoxicity through T cell-mediated attack on hepatic endothelial cells" → add as **REFUTES/QUALIFIES specificity** (on-target off-senescent toxicity). - PMID: 41291258 — "the accumulation of mostly epithelial uPAR-positive cells in the aging gut"

→ add as **SUPPORT** + tissue-dependent target-identity node. - PMID: 40602406 — "Single-cell sequencing in humans confirms uPAR's relevance to senescence and inflammation in RA" → add as the **human single-cell** datapoint. - PMID: 41916312 — "uPAR is broadly expressed in solid tumors enriched for TP53 and RAS pathway mutations" → **QUALIFIES** specificity. - PMID: 41476987 — "the identification of specific and universal senescence markers, potential off-target effects, and long-term safety concerns" → review-level **QUALIFIER**.

Candidate pathophysiology nodes/edges: - Node: *surface uPAR density on senescent cells* (quantitative, tissue-dependent) — not a binary marker. - Edge: *SASP/inflamm-aging → myeloid/endothelial uPAR+ cells* as a parallel, possibly dominant, contributor to therapeutic benefit. - Edge: *anti-uPAR clearance → hepatic-endothelial toxicity* (safety liability).

Candidate ontology terms: senescent epithelial cell (gut), hepatic endothelial cell, neutrophil, monocyte, conventional dendritic cell, preadipocyte; biological processes: SASP, fibrinolysis/plasminogen activation, leukocyte activation.

Candidate status/subtype restrictions: retain **EMERGING**. Add explicit subtype note: *efficacy evidence is mouse-only across metabolic/intestinal/hepatic/arthritic contexts; human evidence restricted to one RA scRNA dataset + a soluble-suPAR aging correlate; no registered human trial as of 2026-10-18.*

Candidate knowledge_gaps / discussion prompts: - "uPAR separates senescent from non-senescent cells only quantitatively; surface fold-change on senescent vs same-type non-senescent cells is unmeasured." - "No human senolytic/uPAR CAR-T trial registered (ClinicalTrials.gov, 2026-10-18)." - "PLAUR human genetics (Open Targets, 2026-10-18) nominate CVD/leukocyte biology, not senescence/aging."

Artifact bundle (`kb/hypotheses/cellular_senescence/upar_senescence_surface_marker_target/openscientist_artifacts`): contains the ClinicalTrials.gov search log, Open Targets associations JSON, HPA expression tables, and SenMayo membership check. The MANIFEST records status/fallback flags and checksummed inputs/outputs. No large raw downloads, controlled data, or credentials are bundled; external sources are recorded by stable identifier.

Proposed Follow-up Experiments / Actions

1. **Run the senescence-resolved surface-uPAR cytometry** (Discriminating Test 1) — the single highest-value experiment; directly tests the specificity premise the KB flags as the open question.
2. **Catalogue uPAR+ cell identity across a cross-tissue single-cell senescence atlas** co-stained for p16/SA-β-gal, to map where uPAR is senescence-enriched vs lineage-driven.
3. **Perform Mendelian randomization** of PLAUR instruments against aging-pace/senescence phenotypes using the Open Targets genetic data already retrieved (F008).
4. **Update the KB** with the curation leads above, preserving EMERGING status and strengthening the specificity qualifier and human-translation caveat.
5. **Monitor ClinicalTrials.gov** for the first registered uPAR/senolytic CAR-T trial; re-run the F007 query set periodically.

Report compiled from 5 OpenScientist iterations: 8 confirmed findings, 33 papers reviewed, 4 computational database analyses (HPA, MSigDB, ClinicalTrials.gov, Open Targets), all executed and logged on 2026-10-18.
